

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285997

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

BUCKWALTER; James et al.

HETEROGENEOUS INTEGRATION OF RADIO FREQUENCY TRANSISTOR CHIPLETS HAVING INTERCONNECTION TUNING CIRCUITS

Abstract

An electronic assembly has a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks. Chiplets are placed in the cavities. At least one chiplet has a second circuit including at least one transistor or switch device and passive tuning circuits including at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network. Electrical interconnects between the chiplets and wafer electrically connect the first circuitry to the second circuitry.

Inventors: BUCKWALTER; James (Santa Barbara, CA), HODGE; Michael (Huntersville, NC), KIM; Justin (San Jose, CA), HERRAULT; Florian (Agoura Hills, CA), GREEN; Daniel (McLean, VA)

Applicant: PseudolithIC, Inc. (Santa Barbara, CA)

Family ID: 90364169

Appl. No.: 19/220027

Filed: May 27, 2025

Related U.S. Application Data

parent US continuation 18491661 20231020 parent-grant-document US 12322714 child US 19220027

parent US continuation 18182314 20230310 parent-grant-document US 11810876 child US 18491661

Publication Classification

Int. Cl.: H01L23/66 (20060101); **H03H7/38** (20060101)

U.S. Cl.:

CPC H01L23/66 (20130101); **H03H7/38** (20130101); H01L2223/6605 (20130101);
H01L2223/6661 (20130101)

Background/Summary

RELATED APPLICATION INFORMATION [0001] This application is a continuation of co-pending patent application Ser. No. 18/491,661, filed Oct. 20, 2023, which is a continuation of patent application Ser. No. 18/182,314, filed Mar. 10, 2023, issued on Nov. 7, 2023 as U.S. Pat. No. 11,810,876, entitled HETEROGENEOUS INTEGRATION OF RADIO FREQUENCY TRANSISTOR CHIPLETS HAVING INTERCONNECTION TUNING CIRCUITS, all of which is incorporated herein by reference.

[0002] A portion of the disclosure of this patent document contains material which is subject to copyright protection. This patent document may show and/or describe matter which is or may become trade dress of the owner. The copyright and trade dress owner has no objection to the facsimile reproduction by anyone of the patent disclosure as it appears in the Patent and Trademark Office patent files or records, but otherwise reserves all copyright and trade dress rights whatsoever.

BACKGROUND

Field

[0003] This disclosure relates to wafers, die, chips and fabrication techniques thereof for electronic assemblies having heterogeneous integration of radio frequencies (RF) transistor chips (e.g., chiplets) within wafer cavities of the wafer and having interconnection tuning circuits for tuning interconnections between the wafer and the chips.

Description of the Related Art

[0004] Electronic assemblies, or hybrid circuits, comprise microelectronic circuits fabricated separately and assembled together to form a single component, which can itself be encapsulated in an electronic circuit package. Assembling microelectronic circuits fabricated separately allows, for example, testing of all the microelectronic circuits separately, prior to assembling them, which, in turn enables improved fabrication yields of the final component. This capability is particularly significant if some of the microelectronic circuits fabricated separately are difficult and/or expensive to manufacture. Assembling microelectronic circuits fabricated separately also allows combining of microelectronic circuits, which themselves employ different materials and different manufacturing processes, into a single final component. This capability can lead to higher circuit performance.

[0005] There exists a need for an electronic assembly using a host wafer having pre-fabricated integrated circuitry, such as passive components, that connect to a microelectronics active chiplet (i.e., with transistors) integrated in a through-wafer cavity of the host wafer. This need may for example be for an assembly for microwave or other radio frequency (RF) integrated circuits that decouple the fabrication of the active circuits (e.g., fabrication of the transistors) from the fabrication of the passive circuitry components (e.g., fabrication of the interconnects, resistors and capacitors). Satisfying this need will allow for much faster manufacturing of the circuits, at lower cost, and a scaling up of active device technologies to circuits without cost and cycle time burden.

Description

DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1A is a schematic top view of a host wafer for heterogeneous integration of radio frequencies (RF) transistor chiplets with tuning circuitry for matching of impedances.

[0007] FIG. 1B is a schematic top view of radio frequencies (RF) transistor chiplets having interconnection tuning circuits for heterogeneous integration into wafer cavities of a host wafer.

[0008] FIG. 1C is a circuit diagram of an electronic assembly having heterogeneous integration of a radio frequencies (RF) transistor chiplet into a wafer, where the chiplet has interconnection tuning circuits.

[0009] FIG. 2A is a schematic diagram of the electronic assembly of FIG. 1C that isolates an example of interconnection tuning circuits having stabilization network circuitry.

[0010] FIG. 2B is a schematic diagram of the electronic assembly of FIG. 1C that isolates an example of interconnection tuning circuits having a neutralization network to improve stability or introduce positive feedback to increase gain of the transistor in the presence of the interconnection to the wafer.

[0011] FIG. 2C is a schematic diagram of an electronic assembly that isolates an example of power delivery to and from chiplets through the interconnection tuning circuits that introduce impedance matching.

[0012] FIG. 2D is a schematic diagram of the electronic assembly of FIG. 1C that isolates an example of interconnection tuning circuits having a negative feedback or degeneration network as is found in low-noise network circuitry to tune the transistor.

[0013] FIG. 2E is a schematic diagram of an electronic assembly that has an example of interconnection tuning circuit that isolates stabilization network circuits of the chiplets in the presence of the interconnect networks.

[0014] FIG. 2F shows circuits having RF transistor chiplets in a wafer, where the chiplets has interconnection tuning circuits.

[0015] FIG. 3A is a schematic cross-sectional view of a device having heterogeneous integration of a radio frequencies (RF) transistor chiplets into a wafer, where the chiplets have interconnection tuning circuits.

[0016] FIG. 3B is a schematic cross-sectional view of a device having heterogeneous integration of a radio frequencies (RF) transistor chiplets onto a wafer, where the chiplets have interconnection tuning circuits.

[0017] Throughout this description, elements appearing in figures are assigned three-digit or four-digit reference designators, where the two least significant digits are specific to the element and the one or two most significant digit may be the figure number where the element is first introduced or fabricated. An element that is not described in conjunction with a figure may be presumed to have the same characteristics and function as a previously-described or subsequently-described element having the same reference designator.

DETAILED DESCRIPTION

Description of Apparatus

[0018] The following describes improved wafers, die, chips and fabrication techniques thereof for electronic assemblies having heterogeneous integration of radio frequencies (RF) transistor chips (e.g., chiplets) within wafer cavities of the wafer and having interconnection tuning circuits for tuning interconnections between the wafer and the chips. The host wafer has circuitry and the radio frequencies (RF) transistor chips within cavities of the wafer, and the chips include one or more types of network circuitry that tune the electrical interconnections between the chips and the wafer. The host wafer can have pre-fabricated interconnects and integrated circuitry, such as passive components, that connect to a chiplet level microelectronics transistor chip integrated in a through-wafer cavity of the wafer. This may form an assembly for integrated circuit devices where the chips contain active circuits from at least one semiconductor technology and the wafers contain passive

(or active) circuits from another semiconductor technology (often a cheaper and larger scale technology). Using a low-cost large-diameter integration platform for the chips with active devices allows for much faster manufacturing of the assembled circuits, at larger scale and lower cost. [0019] The electronic assembled circuit may integrate chiplets having one type of components into a carrier wafer having a different type of components. The electronic assembled circuit may integrate chiplets having high-performance integrated circuits, such as Gallium Nitride (GaN) radio frequency (RF) integrated circuits (ICs) into host wafers having other integrated circuits, such as silicon-based integrated circuits, in a manner that is inexpensive and has high manufacturing yields and short manufacturing cycles. The high performance RF ICs, chips (or chiplets) can have type III-V transistors or other types of transistors and passive circuitry or components, and can be integrated together with resistors, inductors, capacitors and matching networks, as well as active devices from another semiconductor technology into the host wafer. For example, the RF ICs or chiplets can be one type of semiconductor technology that is integrated together with resistors, inductors, capacitors, matching networks, active devices from another semiconductor technology that are part of the host wafer.

[0020] A chiplet may be a chip including the circuitry, material, and/or devices noted herein. It may also be a chip or small chip having active (i.e., transistor) microelectronic devices, active CMOS devices, active microwave IC devices and/or active radio frequency (RF) IC devices. It may also be a chip or small chip having a surface acoustic wave (SAW), bulk acoustic wave (BAW) or other acoustic wave device. A chiplet may have a footprint or top surface area that is half, a third, a fifth or less than a fifth of that of a computer processor chip (e.g., 8086, P3, P4, etc.). Active devices may include active electronic components and/or active electronic circuits.

[0021] The interconnect between the chiplet and wafer maybe mistuned, such as by having AC (alternating current) signal frequencies different at the wafer as compared to the chiplet. The interconnect between the chiplet and wafer maybe mismatched such as by having impedance differences at the wafer as compared to the chiplet that cause instability in AC signals, DC (direct current) signals and/or other electronic signal matching with the input and/or output to the chiplet or a transistor on the chiplet. This lack of tuning may be improved or tuned using the tuning circuits or networks herein.

[0022] FIG. 1A is a schematic top view **100** of a host wafer **110** for heterogeneous integration of radio frequencies (RF) transistor chiplets with tuning circuitry for matching of impedances. Host wafer **110** has back surface **112** and front surface **114** as shown in FIGS. 3A and 3B. Host wafer **110** and/or each cavity **120** has side surfaces **116**, such as a vertical or sidewall surfaces between the back surface **112** and front surface **114**. There may be 3, 4 or more side surface **116**. Typically, there are 4 side surfaces. Wafer capacitors, inductors and resistors may be tuning circuitry for matching of impedances of the interconnects from the chiplet (e.g., see FIG. 1C).

[0023] Host wafer **110** may be or include (e.g., as a mixture of materials or as material layers) silicon, silicon germanium, silicon on insulator, gallium arsenide, indium phosphide, aluminum nitride, diamond, quartz, alumina. If the host wafer **110** only contains interconnections and passive components, it can be a dielectric such as quartz, alumina, or another ceramic. The host wafer **110** may have layers of one or more of these materials in the form of an oxide material, crystalline material and polycrystalline material and/or amorphous material. Host wafer **110** may include some or only passive components such as resistors, capacitors, inductors, through substrate vias, dielectric layers, and/or metal layers (e.g., signal traces or signal planes). Host wafer **110** may include at least one layer of silicon, silicon carbide (SiC), quartz, or another semiconductor wafer material.

[0024] Host wafer **110** may include areas to be diced into integrated circuits, each having passive integrated components (e.g., signal traces, interconnects and conductive vias, resistors, inductors and/or capacitors), a single transistor and/or a plurality of transistors. Silicon is an advantageous choice for host wafer **110**, because it takes advantage of having a lower expense than other

materials; and/or of known microelectronics fabrication processes and of scaling and manufacturing capabilities.

[0025] FIG. 1B is a schematic top view **102** of radio frequencies (RF) transistor chiplets **130** having interconnection tuning circuits for heterogeneous integration into wafer cavities **120** of host wafer **110**. Chiplets **130** have frontside **132** (e.g., a frontside surface) and backside **134** (e.g., a backside surface) as shown in FIGS. 3A and 3B. Each chiplet **130** has side surfaces **136**, such as a vertical or sidewall surfaces between the frontside surface **132** and backside **134**. There may be 3, 4 or more side surface **136**. Typically there are 4 side surfaces. The number of side surfaces **136** of each chiplet **130** may be the same as the number of surfaces **116** of the cavities in the host wafer **110**. Chiplets **130** may each include circuitry described for FIGS. 1C-3B.

[0026] Chiplets **130** may each be or include (e.g., as a mixture of materials or as material layers) silicon (Si), silicon germanium (SiGe), silicon-on-insulator, gallium arsenide (GaAs), indium phosphide (InP), aluminum nitride, quartz, alumina, gallium nitride (GaN). The chiplets **130** may have layers of one or more of these materials in the form of an oxide material, crystalline material and polycrystalline material and/or amorphous material. There may be different electrical component ones or types of chiplets **130** that are manufactured separately from each other. Chiplets **130** can include a GaN, InP or GaAs or any other industry-known electrical component and can be fabricated on a substrate such as Si, SiGe, InP, GaAs, Alumina, or diamond, or any other substrate known in the industry.

[0027] Chiplets **130** or types of chiplets **130** may include transistors used for RF switches, transmit and/or receive circuits; power switches, amplifiers and circuits such as using GaAs, InP, GaN while the host wafer may include transistors such as Si CMOS transistors. The chiplet transistors may have smaller and more expensive electrical components than those of host wafer **110**. There may be one, tens, hundreds, thousands or hundreds of thousands of chiplets **130** embedded in or on one host wafer **110**. There may be between one and one hundred chiplets **130** embedded in one host wafer **110**. Host wafer **110** may have more passive components, lower cost components, routing (e.g., traces, conductive vias and interconnections) than those of chiplets **130**. Host wafer **110** may be fabricated using different microelectronic fabrication techniques or processes than used to fabricate chiplets **130**.

[0028] Chiplets **130** and host wafer **110** can be made of different materials. For example, wafer **110** can be a silicon wafer while chiplets **130** can be a type III-Nitride material component chip. Chiplets **130** may each be or include an integrated circuit having passive integrated components (e.g., signal traces, interconnects and conductive vias, resistors, inductors and/or capacitors), a single transistor and/or a plurality of transistors.

[0029] The chiplets **130**, each include at least one of transistor circuitry and interconnects to contact pads on a frontside **132** of the chiplets **130**. The chiplets **130** may be high-end pre-fabricated active device chiplets that are integrated into host wafer **110** through pick and place assembly on temporary wafer with an adhesive laminate or simply on an adhesive laminate.

[0030] FIG. 1C is a schematic circuit diagram of an electronic assembly **106** having heterogeneous integration of radio frequencies (RF) transistor chiplet **130** into host wafer **110**, where the chiplet **130** has interconnection tuning circuits **140** having passive circuitry. Wafer **110** includes resistors as shown by their schematic symbol one of which is labeled R_w , capacitors as shown by their schematic symbol one of which is labeled C_w , inductors as shown by their schematic symbol one of which is labeled L_w , and grounding as shown by their schematic symbol one of which is labeled GND_w . In some cases, the wafer capacitors and inductors, such as shown by their CLC network under resistor R_w , represent the electrical characteristics of an interconnect such as interconnect **155** and/or **255**. Here, the CLC network may represent an interconnect between the chiplet and the wafer. These characteristics may be part of what is being tuned and/or be part of the interconnection tuning circuitry on the chiplet and/or wafer. The resistor R_w might be considered a separate interconnection tuning circuit on the wafer **110**. The wafer capacitors, inductors and

resistors may be tuning circuitry for matching of impedances of the interconnects from the chiplet. The wafer circuitry may be described as passive (e.g., R, C, L, non-transistor) devices for the purpose of one of tuning or matching networks. Other passive devices may be in or on the wafer. In some cases, the wafer **110** only includes passive, non-transistor devices. In other cases, the wafer circuitry may also include active devices that are not radio frequencies (RF) transistor devices. [0031] Chiplet **130** includes resistors R1, R2 and R3; capacitors C1 and C2; inductor L1, Transistors T1 and T2. Transistor T1 has source S1, drain D1 which is considered the transistor output OUT1 and gate G1 which is considered the transistor input IN1. Transistor T2 has source S2, drain D2 which is considered the transistor output OUT2 and gate G2 which is considered the transistor input IN2. The chiplet circuitry may be described as having both passive (e.g., R, C, L) and active (e.g., at least transistors T1 and T2) devices. Other passive and/or active devices may be in or on the chiplet.

[0032] In some cases, an input signal to the transistor T1 and/or transistor T2; or an input signal to the chiplet such as input INx (e.g., where x is a number as indicated herein); or another input signal to the chiplet also includes a second ground signal, such as GNDr1 or GNDr2 or another ground signal from the chiplet to the transistor or wafer **110**.

[0033] A differential input port between G1/IN1 and G2/IN2; and a differential output port between D1/OUT1 and D2/OUT2 may be used to increase the power generated in a given chiplet area. In this case, S22 may be the “reflection” of output power when Port 2 or OUT2 is connected to a load (not shown). In some cases, an output signal from the transistor T1 and/or transistor T2; or an output signal from the chiplet such as input OUTx (g., where x is a number as indicated herein); or another output signal from the chiplet also includes a second ground signal, such as GNDr1 or GNDr2 or another ground signal to the chiplet or from the transistor or wafer **110**. Here, an input to a transistor or chiplet may have a port 1 between the gate of the transistor and GNDr1; and have an output from the transistor between the drain of the transistor and GNDr1. The input port 1 may have a reflection of input power from a source that is characterized by S11; the output port 2 may have a reflection of output power at the load (e.g., LD1) that is characterized by S22; a forward propagation (gain) of the wave from port 1 to port 2 that is S21; and a reverse propagation (isolation) of the wave from port 2 to port 1 that is S12.

[0034] The available gain GA of the S-parameter network may be characterized by S parameters and the source impedance as:

$$[00001] G_A = \frac{S_{21} (1 - S_{11}^2 - S_{22}^2 - \Delta \Gamma_S)}{1 - S_{11}^2 - S_{22}^2 - \Delta \Gamma_S}$$

[0035] The unilateral gain U may be invariant to a lossless embedding network of any of the networks **140** as:

$$[00002] U = \frac{S_{12} - S_{21}}{\det[1 - SS^*]}$$

[0036] These S-parameters are useful for characterizing how stable a circuit is. A family of stability circles exists over all frequency and input powers. The circuit is only unconditionally stable if the mu u factor remains greater than 1 for all stability circles possible operating conditions, such as where:

$$[00003] \mu = \frac{1 - |S_{11}|^2}{|S_{22} - S_{11}^* S_{12}| + |S_{12} S_{21}|} > 1.$$

[0037] That is any of the networks **140**, without circuitry of the wafer **110** (except with the network of FIG. 2C of inductors L3-L8 on the wafer) may make the transistor or chiplet unconditionally stable by causing u to be greater than 1 under operating conditions, such as when the transistor or chiplet is amplifying or passing an RF signal.

[0038] Each of transistors T1 and T2 may be a radio frequencies (RF) transistor device. Each of transistors T1 and T2 may operate at (e.g., amplify, pass and/or switch) radio frequencies and/or be an RF wave transistor device. Each may operate on electronic signals with frequencies between 10 MHz and 300 GHz, 30 GHz and 300 GHz or 10 MHz and 10 GHz. Each of transistors T1 and T2

may be a millimeter-wave or a microwave transistor device. Each of transistors **T1** and **T2** may be a single ended RF amplifier, such as used to make a chain of amplifiers, a receiver using amplifiers, a mixer, a switch or a frequency multiplier. Each of transistors **T1** and **T2** may be part of a differential amplifier, placed in parallel with a shared drain and source but distinct gate or base connections for a frequency multiplier, or in a series connection of devices for a cascode. Additionally, transistors **T1** and/or **T2** might be part of a switch pair of transistors for an RF switch or mixer. Four total transistors might be used in a single chiplet for a FET ring mixer. Each of transistors **T1** and **T2** may be a heterojunction bipolar transistor (HBT), in which case they will have a base, emitter and collector instead of a gate, source and drain, respectively. Each of transistors **T1** and **T2** may be used in common gate mode by grounding both transistor's gates and using the sources as inputs. Each of transistors **T1** and **T2** may be used with the input at the source and an output at the drain.

[0039] In the example of FIG. 1C, transistor **T1** is an RF amplifier and has a signal phase output of a differential amplifier, and transistor **T2** is another RF amplifier and has a signal phase output of the differential amplifier that is 180 degrees out of phase with the signal phase of transistor **T1**. Transistors **T1** and **T2** may be two differential RF transistors and there may be interconnects from these transistors to contact pads **138** on a frontside surface of the chiplet **130**.

[0040] Wafer **110** may also include circuit networks to provide a virtual ground at GNDr1 such as an AC ground and circuitry to apply the DC bias voltage applied to the node **N1** between resistors **R1** and **R2**. Wafer **110** may also include GNDr2 such as an AC ground and DC bias voltage applied to one end of resistor **R3** that is opposite the node **N1** between resistors **R1** and **R2**.

[0041] Assembly **106** also includes interconnects **155** between contacts **118** of wafer **110** and contacts **138** of chiplet **130**. A contact may be a contact pad or other electrical connection to circuitry of the wafer or chiplet. The interconnect **155** may be a trace (e.g., of a PCB), a wire or another electrically conductive connection between the contacts. The electrical interconnects **155** electrically connect the wafer circuitry to the chiplet circuitry. There may be similar interconnects and contact pair at the border (e.g., at sidewalls) between the wafer and chiplet for each electrical connection needed at that border.

[0042] The electrical interconnects **155** electrically connect the circuit devices of the wafer and the transistors of the chiplets. The interconnect between the chiplet and wafer maybe mistuned, such as by having mismatches in stability, AC signal, DC signal, impedance, frequency and/or other electronic signal matching with the input and/or output to the chiplet or a transistor on the chiplet. This lack of tuning may be improved or tuned using the tuning circuits **140**. Notably, the interconnection tuning circuits **140** reduce this electrical mistuning of passive devices and the transistors caused by the electrical interconnects that may impact circuit performance and reduces the uncertainty of the electrical impedance and frequency of signals passed across the interconnects to improve the circuit performance.

[0043] The interconnection tuning circuits **140** may be any or all of the resistors, inductors and capacitors shown on the chiplet **130** or other chiplets herein. In some cases, it may also include inductors on the wafer such as shown in FIG. 2C. The interconnection tuning circuits **140** may include at least one of a stabilization network (e.g., network of electronic circuitry), a gain boosting network, a power delivery network, and/or a low-noise network. Each of these networks may be an embedded network that is embedded into wafer **110** as part of the chiplet.

[0044] There may be more or fewer electrical and/or circuit components than shown in FIG. 1C for chiplet **130**, wafer **110**, and/or interconnection tuning circuits **140**. Although only one is shown in FIG. 1C, there may be multiple chiplets **130** in cavities of wafer **110**. There may be between 10 and 1000 such chiplets in the wafer.

[0045] FIG. 2A is a schematic diagram of the electronic assembly **106** of FIG. 1C that isolates an example of stabilization network circuitry **142** of the interconnection tuning circuits **140**. Circuitry **142** may be a network of electronic circuitry having passive circuitry that includes at least one of

resistors R1, R2 and/or R3. Circuitry 142 may include only resistors R1 and R2. In another case, circuitry 142 includes all of resistors R1, R2 and R3. The R3 resistor is introduced to provide common-mode stability or to provide feedback to the virtual ground on the drain/output circuitry. Circuitry 142 may optionally include ground GNDr1 and/or GNDr2. Circuitry 142 may optionally include one or more of inductor L1, capacitor C1 and/or capacitor C2. The stabilization network circuitry 142 may be stabilization network circuit for transistor T1 and/or T2 in spite of the electrical characteristic or mistuning of interconnection circuit 155 and/or 255. Additionally, the resistor R3 may not grounded to GNDr1 but may instead connect to the drain D1 and/or D2 of the transistor T1 and/or T2 through some external circuit that exists in the wafer. In some cases, resistors R1, R2 and R3 are not grounded to ground GNDr1 or GNDr2 (although they may be grounded to a “virtual” ground), and are used to provide a DC bias to the gates G1 and G2 while not influencing the differential circuit's RF frequency performance. The ground GNDr1 or GNDr2 may provide DC bias voltage through resistors R1 R2 and R3 to the gates G1 and G2. Thus, the resistors R1 R2 and R3 provide stabilization of the DC bias to the gates G1 and G2, such as by setting the DC bias on the gates of the transistors.

[0046] In some cases, circuitry 142 may include at least one resistor R1 between the input IN1 of transistor T1 and the node N1 to ground GNDr1; at least one resistor R2 between the input IN2 of transistor T2 and the node N1; and at least one resistor R3 between the node N1 and ground GNDr2. Here, ground GNDr1 may be a first AC ground that allows a DC voltage that provides a desired first DC bias to the inputs IN1 and IN2; and GNDr2 may be a second AC ground and DC voltage that provides a desired second DC bias to the inputs the inputs IN1 and IN2.

[0047] In some cases, circuitry 142 may include at least one resistor R1 (and optionally R2) between the input IN1 and the input IN2; and at least one resistor R2 (and optionally R1) between the input IN2 and the input IN1, as shown in FIG. 1C. There may optionally be capacitors C1 and C2 between these inputs and outputs as shown in FIG. 1C. There may optionally be inductor L1 between the S1 and S2 as shown in FIG. 1C.

[0048] Any one, two or all 3 of these resistors R1, R2 and R3 may provide desired DC bias and common mode stability to the inputs IN1 and IN2 as compared to without the resistor(s). Any one, two or all 3 of these resistors R1, R2 and R3 may provide desired impedance matching at the input and output of each transistor T1 and T2 as compared to without the resistor(s).

[0049] In some cases, these resistors R1 and R2 may provide desired differential mode (e.g., where the operation or output of transistor T2 is 180 degrees out of phase with transistor T1) stability to the inputs IN1 and IN2 as compared to without those resistors; and may provide desired impedance matching at the input and output of each transistor T1 and T2 as compared to without those resistors. This stability with reduce unwanted changes in power levels, output of noise and/or oscillation of the transistor outputs. Here, resistors R1 and R2 may provide desired shunt resistance to ground (e.g., GNDr1) that is used for the stabilization.

[0050] In some cases, the resistor R3 (and optionally a capacitor such as C1 in series with resistor R3) may provide desired common mode (e.g., where the operation or output of transistor T2 is 0 degrees out of or in phase with transistor T1) stability to the inputs IN1 and IN2 as compared to without those resistors; and may provide desired common mode impedance matching at the input and output of each transistor T1 and T2 as compared to without those resistors. This stability with reduce unwanted changes in power levels, output of noise and/or oscillation of the transistor outputs. Here, resistor R3 may provide desired series feedback (e.g., to components of the chiplet or wafer electrically connected to the transistor output) that is used for the common mode stabilization.

[0051] The DC stability and/or impedance matching produces an unconditionally stable chiplet with respect to signal received on the interconnects 155 from the wafer and thus allows the active device transistors T1 and T2 of chiplet 130 to also be unconditionally stable. It is considered that stabilization of the transistors T1 and T2 might incorporate lossy components such as resistors R1,

R2 and R3 and reactive components such as capacitors C1 and C2 and inductor L1 to stabilize the transistors T1 and T2 in a specific frequency band of operation as noted for the transistors. The stabilization can be applied to a single transistor but also to a differential pair example of transistors T1 and T2. In this case, both the stability of the differential and common-modes would be guaranteed through the network circuitry 142 implemented on the active chiplet.

[0052] FIG. 2B is a schematic diagram of the electronic assembly 106 of FIG. 1C that isolates an example of gain boosting network circuitry 144 of the interconnection tuning circuits 140.

Circuitry 144 may be a network of electronic circuitry having passive circuitry that includes at least one of capacitors C1 and/or C2. Circuitry 144 may include both of capacitors C1 and C2. Circuitry 144 may optionally include ground GNDr1 and/or GNDr2. Circuitry 142 may optionally include one or more of resistors R1, R2 and/or R3, such as shown in FIG. 1C. Circuitry 144 may be a neutralization network to improve stability of transistor T1 and/or T2; or to introduce positive feedback to transistor T1 and/or T2 to increase gain of the transistors in the presence of or through the interconnections 155 and/or 255 to the wafer.

[0053] In some cases, circuitry 144 may include at least one capacitor C1 between the input IN2 of transistor T2 and the output OUT1 of transistor T1; and at least one capacitor C2 between the input IN1 of transistor T1 and the output OUT2 of transistor T2. These capacitors counter-act the parasitic gate-drain capacitance of transistors T1 and T2.

[0054] In some cases, circuitry 144 may include at least one resistor R1 and R2 between the input IN1 and input IN2 as shown in FIG. 1C. There may optionally be inductor L1 between the S1 and S2 as shown in FIG. 1C.

[0055] Both of these capacitors C1 and C2 may provide desired gain boosting and/or a “neutralization”—such as electrically isolating the input IN from the output OUT of each transistor T1 and T2 of the chiplet as compared to without the capacitor(s). Any one or both of these capacitors C1 and C2 may provide desired impedance matching of the impedance the transistor sees at its input IN and output OUT from the wafer 110 as compared to without the capacitor(s). In some cases, the network circuit 144 implemented on the active chiplet 130 can realize desired specific optimization for gain of the transistors T1 and T2. For instance, both of these capacitors C1 and C2 may provide desired unilateralization to improve the gain of the transistor(s) by eliminating the reverse signal flow of the transistor(s) (e.g., cause $S_{12}=0$) as compared to without the capacitor(s). In other words, each of transistors T1 and T2 can be unilateralized such that the reverse isolation (S_{12} —backwards transfer function from OUT to IN) is zero. Any one or both of these cross-coupled capacitors C1 and C2 may provide a desired negative capacitance to cancel C_{gd} (the capacitance between the gate and drain) of the transistor(s) as compared to without the capacitor(s). When operating in differential mode, where transistor T2 is 180 degrees out of phase with transistor T1, the cross coupled capacitors C1 and C2 act as a negative capacitor to the transistors T1 and T2, thus canceling the C_{gd} of each respective transistor as compared to without the capacitors. Another gain boosting technique or circuit (not shown) can be used to increase the gain beyond the unilateral gain. The limit of either or both of these gain boostings, while remaining unconditionally stable, is to reach $4U$ (e.g., $4\times$ the maximum unilateral gain of the T1 and T2).

[0056] In other embodiments, any one or both of these capacitors C1 and C2 may be replaced with an inductor such as an inductor as noted herein between the gate of one of transistor T1 and T2 and the drain of the other one of one of transistor T1 and T2.

[0057] FIG. 2C is a schematic top view of an electronic assembly 200 that isolates an example of power delivery circuits 145, 146 and 147 of the interconnection tuning circuits 140 having passive circuitry of chiplets 133, 134 and 135, each having active transistor T1. In some cases, one of power delivery and efficiency circuits 145, 146 or 147 is also on the chiplet having active transistor T2, such as where chiplets 133, 134 and 135 have both transistor T1 as shown in FIG. 2C and also have transistor T2 (e.g., see assembly 106) with one of circuits 133, 134 or 135. Each of circuits 145, 146 and 147 may be an example of a power delivery circuit to and from chiplets 133, 134 and

135 through the interconnection networks including through interconnections **155** and/or **255** that introduce impedance matching through interconnections **155** and/or **255**, such as from the wafer to transistor **T1**.

[0058] There may be more or fewer electrical and/or circuit components than shown in for chiplets **133**, **134** and **135**, wafer **110**, and/or network circuits **133**, **134** or **135**. The circuitry of chiplets **133**, **134** and **135** may be described as having both passive (e.g., R, C, L) and active (e.g., at least transistor **T1**) devices. Other passive and/or active devices may be in or on the chiplet. Although only three chiplets are shown, there may be numerous chiplets in cavities of wafer **110**. There may be between 10 and 1000 such chiplets in the wafer.

[0059] Each of chiplets **133**, **134** and **135** may represent a version of chiplet **130**. Each of circuits **145**, **146** or **147** may represent a version of circuitry **140**.

[0060] Interconnects **255** are examples of interconnects such as those described by contact **118**, interconnect **155** and contact **138**. The electrical interconnects **255** electrically connect the wafer circuitry to the chiplet circuitry. There may be interconnects similar to interconnects **255** at the border (e.g., at sidewalls) between the wafer and chiplet for each electrical connection needed at that border.

[0061] In some cases, circuitry **145**, **146** or **147** may include at least one capacitor **C5** or **C7** between the gate **G1** of transistor **T1** and the ground **GNDr1**; and/or at least one capacitor **C4** or **C6** between the drain **D1** of transistor **T1** and the output **OUT6** or **OUT 7** of chiplets **133** and **134**.

[0062] Circuitry **145** may be a network of electronic circuitry that includes at least two capacitors **C3** and **C4** and two inductors **L2** and **L3**. Inductor **L2** is connected between gate **G1** and ground **GNDr1** and/or other active or passive circuitry of the wafer **110**. Inductor **L3** is connected between the load **LD1** and the node between drain **D1** and capacitor **C4**. Capacitor **C3** is connected between the input **IN6** and the node between the inductor **L2** and gate **G1**. Capacitor **C4** is connected between the drain **D1** and the output **OUT6**. Inductor **L2** may be used to provide DC bias to gate **G1** and inductor **L3** may be used to provide DC bias to drain **D1**. Capacitors **C3** and **C4** may be used to isolate the transistor from another part of the circuit and/or prevent DC current from flowing into a termination port.

[0063] Load **LD1** may be a 50-Ohm load which is a convention used in the definition of S-parameters that describe power flow into and out of a transistor.

[0064] In the case shown, inductors **L2** and **L3** are on wafer **110**, such as due to their larger size than the capacitors. In other cases, inductors **L2** and **L3** are on chiplet **133**.

[0065] Inductors **L2** and **L3**; and capacitors **C3** and **C4** may create an electrical network that increases power delivery and efficiency for transistor **T1** (e.g., at output **OUT6**), by transforming the impedance of the source and/or load to an impedance seen at the transistor that maximizes the power delivered or the power added efficiency, as compared to without the inductors and capacitors. As an impedance transformation network, any of circuits **145**, **146** and **147**, is an electrical network that takes one impedance (**Z1**) at one port (such as the input **IN** or output **OUT** of that circuit) and transforms it to another impedance (**Z2**) at a second port (such as the other of the input **IN** or output **OUT** of that circuit). For transistor **T1**, they may maximize output power (e.g., at output **OUT6**) under mismatches in the load impedance (e.g., of **LD1**) or introduce load modulation for power efficiency under different output power conditions, as compared to without the inductors and capacitors.

[0066] Circuitry **146** may be a network of electronic circuitry that includes at least two capacitors **C5** and **C6** and two inductors **L4** and **L5**. Capacitor **C5** is connected between gate **G1** and ground **GNDr1**. Inductor **L5** is connected between the load **LD1** and the node between drain **D1** and capacitor **C6**. Inductor **L4** is connected between the input **IN7** and the node between the capacitor **C5** and gate **G1**. Capacitor **C6** is connected between the drain **D1** and the output **OUT7**. Inductor **L4** may be used to provide DC bias to gate **G1** and inductor **L5** may be used to provide DC bias to drain **D1**. Capacitors **C5** and **C6** may be used to isolate the transistor from another part of the

circuit and/or prevent DC current from flowing into a termination port.

[0067] In the case shown, inductors **L4** and **L5** are on wafer **110**, such as due to their larger size than the capacitors. In other cases, inductors **L4** and **L5** are on chiplet **134**.

[0068] Inductors **L4** and **L5**; and capacitors **C5** and **C6** may provide desired increased power delivery and efficiency for transistor **T1** (e.g., at output **OUT7**) through impedance transformations, as compared to without the inductors and capacitors. For transistor **T1**, they may maximize output power (e.g., at output **OUT7**) under mismatches in the load impedance (e.g., of **LD1**) or introduce load modulation for power efficiency under different output power conditions, as compared to without the inductors and capacitors.

[0069] Circuitry **147** may be a network of electronic circuitry that includes at least one capacitor **C7** and three inductors **L6**, **L7** and **L8**. In some cases, capacitor **C7** is on the host wafer **110** and not within chiplet **135**. Inductor **L7** is connected between the load **LD1** and the node between drain **D1** and inductor **L8**. Inductor **L6** is connected between the input **IN8** and the node between the capacitor **C7** and gate **G1**. Inductor **L8** is connected between the drain **D1** and the output **OUT8**.

[0070] In the case shown, inductors **L6-8** are on wafer **110**, such as due to their larger size than the capacitors. In other cases, inductors **L6-8** are on chiplet **135**.

[0071] Inductors **L6**, **L7** and **L8**; and capacitor **C7** may provide desired tuning that increases power delivery and efficiency for transistor **T1** (e.g., at output **OUT8**) through impedance transformations, as compared to without the inductors and capacitors. For transistor **T1**, they may maximize output power (e.g., at output **OUT8**) under mismatches in the load impedance (e.g., of **LD1**) or introduce load modulation for power efficiency under different output power conditions, as compared to without the inductors and capacitors. By moving the inductors to the host wafer **110**, substantial reduction in cost and time is achieved. In some cases, it is less expensive and results in fewer defects when the inductors are formed on the wafer **110** instead of on a chiplet because the wafer is a cheaper and easier to process semiconductor material.

[0072] Power delivery and efficiency circuits **145**, **146** and **147** may also be used on either or both of transistors **T1** and/or **T2** as shown in FIG. **1C**.

[0073] FIG. **2D** is a schematic top view of the electronic assembly **106** of FIG. **1C** that isolates an example of low-noise network circuitry **148** of the interconnection tuning circuits **140**. Circuitry **148** may be a network of electronic circuitry having passive circuitry that includes at least one inductor **L1**. Circuitry **148** may include only inductor **L1**. In another case, circuitry **148** includes all of inductor **L1** and capacitor **C1** and/or capacitor **C2**. Circuitry **148** may be an interconnection tuning circuit having negative feedback or be a degeneration network through interconnects **155** and/or **255** as is found in low-noise network circuitry to tune the transistor **T1** and/or **T2**.

[0074] In some cases, circuitry **148** may include at least one inductor **L1** between the source **S1** of transistor **T1** and the source **S2** of transistor **T2**. This inductor may have a value between 5 pH and 500 pH.

[0075] In some cases, circuitry **148** may also include at least one capacitor **C1** between the input **IN2** of transistor **T2** and the output **OUT1** of transistor **T1**; and at least one capacitor **C2** between the input **IN1** of transistor **T1** and the output **OUT2** of transistor **T2**.

[0076] Inductor **L1** may provide desired tuning for low-noise operation that simultaneously matches the transistor for low noise and/or gain matching

[0077] Inductor **L1** can act as a local “ac ground” that avoids a ground current from flowing through the interconnects. By defining a ground on **148**, the interconnect will not impact the differential behavior of the circuitry.

[0078] In other embodiments, inductor **L1** may be replaced with a capacitor such as an capacitor **C1** the sources of transistor **T1** and **T2**. In other embodiments, inductor **L1** may be replaced with an AC ground such as ground **GNDr1** or **GNDr2**.

[0079] In some cases, to accurately apply transistor models, the high-frequency (AC) ground of the chiplets must be established, such as by **GNDr1** and/or **GNDr2**. Heterogeneous integration adds

parasitics that change the AC ground. A key feature of the networks **140** is that they can use differential circuitry in the chiplet to establish a local ground. A reactive or resistive source connection network, such as any of the networks **140**, creates a well-defined AC ground.

[0080] FIG. **2E** is a schematic top view of an electronic assembly **250** that isolates an example of stabilization network circuits **241**, **242** and **243** of the interconnection tuning circuits **140** having passive circuitry of chiplets **230**, **231** and **232**, each having active transistor **T1**. There may be more or fewer electrical and/or circuit components than shown in for chiplets **230**, **231** and **232**, wafer **110**, and/or stabilization network circuits **241**, **242** and **243**. The circuitry of chiplets **230**, **231** and **232** may be described as having both passive (e.g., R, C, L) and active (e.g., at least transistor **T1**) devices. Other passive and/or active devices may be in or on the chiplet. Although only three chiplets are shown, there may be numerous chiplets in cavities of wafer **110**. There may be between 10 and 1000 such chiplets in the wafer. Each of circuits **241**, **242** and **243** may be an isolation stabilization network circuit of chiplets **230**, **231** and **232**, respectively, in the presence of the interconnect networks extending through interconnects **155** and/or **255**.

[0081] Each of chiplets **230**, **231** and **232** may represent a version of chiplet **130**. Each of circuits **241**, **242** and **243** may represent a version of circuitry **140**.

[0082] Circuitry **241** may be a network of electronic circuitry that includes at least one resistor **R4** between gate **G1** and ground **GNDr1**. Circuitry **241** includes input **IN3** electrically attached to a node between gate **G1** and resistor **R3**; output **OUT3** at the drain **D1**; and a load **LD1** at a node between drain **D1** and output **OUT3**.

[0083] Resistor **R4** may provide desired DC bias and DC stability to the input **IN3** as compared to without the resistor. Resistor **R4** may provide desired impedance matching at the input **IN3** and output **OUT3** of transistor **T1** as compared to without the resistor.

[0084] Circuitry **242** may be a network of electronic circuitry that includes at least one resistor **R5** in series with one capacitor **C8** between gate **G1** and output **OUT4**. Although the resistor is shown closer than the capacitor to the gate, the order of those components may be reversed. Circuitry **242** includes input **IN4** electrically attached to a node between gate **G1** and resistor **R5**; output **OUT4** at the drain **D1**; and a load **LD1** at a node between drain **D1**, an end of capacitor **C8** and output **OUT4**.

[0085] Resistor **R5** and capacitor **C8** may provide stabilization over a range of frequencies between the input **IN4** and/or output **OUT4** as compared to without the resistor and capacitor. Resistor **R5** and capacitor **C8** may provide desired DC and/or AC impedance matching at the input **IN4** and output **OUT4** of transistor **T1** as compared to without the resistor and capacitor. Resistor **R5** and capacitor **C8** may provide series RC feedback between the gate and drain compared to without the series RC feedback in terms of DC and/or AC biasing.

[0086] In some cases, the electrical connection between output **OUT4** and the node between capacitor **C5** and load **LD1** may include passive or active devices.

[0087] Circuitry **243** may be a network of electronic circuitry that includes at least one resistor **R6** between output **OUT5** and load **LD1**. Circuitry **243** includes input **IN5** electrically attached to gate **G1**; output **OUT5** at a node between the drain **D1** and output **OUT5**; and a load **LD1** at the end of the resistor **R6** away from the node between drain **D1** and output **OUT5**.

[0088] Resistor **R6** may provide desired DC bias and DC stability to the output **OUT5** as compared to without the resistor. Resistor **R6** may provide desired impedance matching at the input **IN5** and output **OUT5** of transistor **T1** as compared to without the resistor.

[0089] The DC stability and/or impedance matching of stabilization network circuits **241**, **242** and **243** produces unconditionally stable chiplets **230**, **231** and **232** with respect to signals received on the interconnects **255** from the wafer **110** and thus allows the active device transistor **T1** to also be unconditionally stable. It is considered that stabilization of the transistor **T1** might incorporate additional lossy components such as resistors and reactive components such as capacitors and/or inductors to stabilize the transistor **T1** in a specific frequency band of operation as noted for the transistors. This stabilization can be applied to a single transistor but also to a differential pair of

transistors. In this case, both the stability of the differential and common-modes would be guaranteed through the network circuits **241**, **242** and **243** implemented on the active chiplets **230**, **231** and **232**.

[0090] Stabilization network circuits **241**, **242** and **243** may also be used on either or both of transistors **T1** and/or **T2** as shown in FIG. **1C**. Any network circuitry of circuitry **140** may be an impedance matching network.

[0091] FIG. **2F** shows circuits **260-269** having transistor **T1** and/or **T2** as RF transistors; with input **IN**, output **OUT**, AC ground shown by the triangular ground symbol (which may also be a DC voltage bias), load **LO**; switch control signal **S**; and input output **I/O**.

[0092] FIG. **2F** shows circuits **260** and **261** having transistor **T1** as a single-ended RF amplifier with an input at the gate and source, respectively, and an output at the drain.

[0093] FIG. **2F** also shows circuit **262** having transistors **T1** and **T2** in a series connection of devices for an RF cascode with an input at the gate of transistor **T1**, and an output at the drain of transistor **T2**.

[0094] Next, FIG. **2F** also shows circuit **263** having both of transistors **T1** and **T2** as a differential RF amplifier with inputs at the gates and outputs at the drains of each transistor.

[0095] Now, FIG. **2F** also shows circuit **264** having both of transistors **T1** and **T2** as a frequency multiplier with inputs at the gates of each transistor and an output tied to the drains of both transistors.

[0096] In addition, FIG. **2F** also shows circuit **266** having two sets of transistors **T1** and **T2** as four total transistors on a single chiplet as an RF FET ring mixer or provider of frequency translation with two inputs **IN1** and **IN2**; two outputs **OUT1** and **OUT2**; and two loads **LO1** and **LO2**.

[0097] Then, FIG. **2F** also shows circuit **267** having transistor **T1** as an RF switch transistor for an RF switch or mixer with an input at the source, a switch control signal **S1** at the gate and an output at the drain.

[0098] Also, FIG. **2F** also shows circuit **268** having transistors **T1** and **T2** in series as RF switch transistors for an RF switch or mixer with an input output **I/O** at the drain of each transistor and a switch control signal at the gate of each transistor.

[0099] Finally, FIG. **2F** also shows circuit **269** having transistors **T1** and **T2** as part of RF switch transistors for an RF switch or mixer with an input at the source of transistor **T1**; an output at the drains of transistors **T1** and **T2**; a switch control signal at the gate of each transistor.

[0100] Any of tuning circuitry **140-148** and/or **241-243**, such as shown in FIGS. **1C** and **2A-2E**, may be applied to any of circuits **260-269**. Chiplets that exhibit gain as illustrated in **260-263** require stabilization networks to ensure robust operation in the presence of interconnection to the tuning circuits. Chiplets that generate power as illustrated in **260-264** are improved in performance by tuning networks as indicated by the sample in **140-148** and/or **241-243** that produce desirable fundamental or harmonic tuning networks. Switching networks including mixers and RF switches indicate in **265-269** benefit from the network techniques in FIG. **1C** to prevent signal power loss on the chiplet and, potentially, due to the interconnection networks.

[0101] FIG. **3A** is a schematic cross-sectional view of a device **300** having heterogeneous integration of a radio frequencies (RF) transistor chiplets **130** into a wafer **110**, where the chiplets have interconnection tuning circuits **140**. Device **300** may include the one or more of the devices of any or all of FIGS. **1A-2F**.

[0102] Device **300** may be an electronic assembly having an encapsulation material layer **370** having a top surface **372** and a back surface **374**. Device **300** has a host wafer **110** having back surface **112** and front surface **114**, with the back surface **112** of the wafer bonded to the top surface **372** of an encapsulation material layer **370** except for cavities **120** in the wafer **110** formed over a plurality of areas **376** of the top surface **372**. The cavities may extend from back surface **112**, through the wafer and to front surface **114**. The cavities have side surfaces **116**. The back surface **112** of the wafer may be directly attached to and touching the top surface **372**. The bond between

the back surface **112** and the top surface **372** may be a covalent, chemical or atomic bond.

[0103] Chiplets **130** have a backside **134** and a frontside **132**, with the backsides **134** of the chiplets **130** bonded directly to at least portion **378** of the plurality of areas **376** of the top surface **372** of the encapsulation material layer. Portion **378** may be the footprint of the chiplet **130** on top surface **372** within the cavity **120**. A gap **350** between side surfaces **116** and **136** may be the difference between area **376** and portion **378**. The backside **134** may be directly attached to and touching the top surface **372**. The bond between the backside **134** and the top surface **372** may be a covalent, chemical or atomic bond.

[0104] The cavities **120** may be through-substrate holes or through substrate holes etched in the wafer at the areas **376**. The, chiplets **130** may be embedded into the wafer **110** at the substrate holes or at cavities **120**.

[0105] A lateral material **360** extends between side surfaces **136** of the chiplets **130** and the side surfaces **116** of the wafer or cavities. The lateral material **360** may mechano-chemically bond the side surfaces **136** of the chiplets **130** to the side surfaces **116** of the wafer. The lateral material **360** may form a mechanical and/or a chemical bond to the side surfaces **136** and to the side surfaces **116**. In some cases, the lateral material **360** is a molded material and the bonding is a mechano-chemical bond.

[0106] Material **360** may be a dielectric material. Material **360** may not be a metal and may be an electrical insulator. Material **360** may be or include material that is not conductive, is may not be a semiconductor, may be a plastic, may not be an alloy, may be a bio-material. Material **360** may be an epoxy. It may be an electrical insulator epoxy with electrical insulator particles. It may be epoxy with silica or SiO₂ particles.

[0107] In other cases, the lateral material **360** is not a dielectric material. In these cases, the lateral material **360** may be a metal, a conductor, an alloy or a semiconductor. Material **360** may be an epoxy. It may be a conductive epoxy with conductive particles. It may be epoxy with a metal particles. In these cases, there is a dielectric layer or a space (e.g., of air) between the material **360** and the interconnects **155** and/or **255**.

[0108] The lateral material **360** is disposed in gaps **350** between the side surfaces **136** of each of the chiplets **130** and the side surfaces **116** of the corresponding wafer cavity that each chiplet **130** is disposed in. The gap **350** has a width gw of between $\frac{1}{5}$ (one fifth) and 10 times a thickness tw of the wafer **110** or chiplets **130**.

[0109] The thickness tw of the wafer may be between 20 and 200 microns. It may be between 50 and 125 um. It may be 75 um. The thickness of one, many or all of the chiplets may be that same as that of the wafer.

[0110] A thickness te of the encapsulation material layer may be between 3 and 100 microns. It may be between 5 and 25 microns. It may be 15 um.

[0111] Each of the chiplets **130** have between 3 and 6 sides. They may have 4 sides. The sides may be straight, curved or wavy in profile as viewed from a top perspective. The cavities **120** may have the same number of and sides corresponding to the shapes of the sides of the chiplets **130**.

[0112] The encapsulation material layer **370** may be a high-thermal-conductivity backside metallization layer that improves heat transfer from the chiplets **130** to the wafer **110**. Layer **370** may be a thermal plane that improves heat conduction away from the chiplets by increasing thermal conduction from the chiplets **130** and to layer **370** and/or wafer **110**. Layer **370** be a material in direct contact with the chiplets **130** to increase thermal conduction between the materials of the chiplets **130** and that of layer **370**. In some cases, the encapsulation material layer **370** has a coefficient of thermal expansion between those of or equal to one of those of the wafer **110** and of the chiplets **130**.

[0113] In other cases, there is no encapsulation material layer **370** and the chiplets **130** are bonded to the wafer **110** using the lateral material. the chiplets **130** may be bonded to the wafer **110** using only the lateral material.

[0114] Interconnects **155** may be formed directly on or may be formed over (e.g., formed on a dielectric or air gap over) the lateral material **360** and connect electrical (e.g., power, ground and/or signal) contacts **138** of the chiplets **130** to contacts **118** of the wafer **110**. Interconnects **155** may include direct interconnect routing or traces that is formed directly on the lateral material (e.g., without any dielectric/air gap), and that extends from the chiplet contacts **138** to wafer contacts **118** and electrical routing. The interconnect routing **155** may include low loss high-performance DC, RF, and mm-wave routing from the chiplet contacts **138**, directly on the lateral material **360**, and to wafer contacts **118**. Interconnects **155** may be directly on material **360** by being bonded to and/or directly attached to (e.g., touching) the top surface of the lateral material **360**.

[0115] In some cases, wafer **110** includes an electronic integrated circuit (not shown), at least one integrated circuit contact **118** (e.g., contact pad) formed on the front wafer surface **114**, and at least one through-wafer cavity **120** having side surface **116** that join back surface **112** to front surface **114**. In some cases, a chiplet **130** is held in the through-wafer cavity **120** by a lateral material **360** that attaches at least one side surface **116** of the through-wafer cavity **120** to at least one side surface **136** of the chiplet **130**. In some cases, lateral material **360** fills gap **350** of the cavity, thus attaching most of the side surfaces **136** of chiplet **130** to the side surfaces **116** of through-wafer cavity **120**; however material **360** does not attach the backside **134** of chiplet **130** to top surface **372** of layer **370**.

[0116] A passivation layer (not shown) can be arranged on most of the front surface **114** of wafer **110**. Conducting vias (e.g., TWVs) arranged through the passivation layer can connect the active and/or passive circuitry of wafer **110** to contacts **118** (e.g., contact pads) on front surface **114**. Wafer **110** can be a silicon wafer or substrate, which allows taking advantage of known fabrication processes and manufacturability on large wafer diameters.

[0117] It is noted that wafer **110** can include any integrated circuit, active or passive, made possible by a chosen manufacturing process; for example, a CMOS manufacturing process. In some cases, the thickness of the one or more integrated circuit layers can for example be only a fraction of the thickness t_w of wafer **110** (for example between 1/10 and 1/1000 of the thickness of wafer **110**; for example 50 nm thick with a wafer 50 μm thick). In some cases, the thickness of wafer **110** can be reduced after fabrication of integrated circuits of the wafer and for example before etching the through-wafer cavity **120** or after filling gap **350** with lateral material **360**.

[0118] Chiplet **130** may include one or more transistors having its terminals connected to at least one integrated circuit contact **138** (e.g., contact pad), such as by a conductive via (not shown). Chiplet **130** can comprise a substrate and integrated circuit layers formed on top of its substrate, the thickness of the integrated circuit layers being for example only a fraction of the thickness of the substrate (for example between 1/10 and 1/1000 of the thickness of the substrate). In some cases, the total thickness of chiplet **130** is smaller than the total thickness of host wafer **110**. In some cases, lateral material **360** contacts the side surfaces **136** of chiplet **130** along most of their height (at least 50% of the height, starting from close to the top surface of chiplet **130**). Preferably, lateral material **360** contacts essentially all of the side surfaces **136** of chiplet **130**. Preferably, lateral material **360** fills completely gap **350**, up to a level essentially flush with the front surface **114** of host wafer **110**.

[0119] In some cases, lateral material **360** and/or layer **370** holds the chiplet **130** such that the chiplet frontside **132** is flush with the front surface **114**. Being “flush” may be understood as meaning that the two surfaces are in a same plane, or have, with respect to each other, a small or negligible height difference. The two surfaces may be flush, such as resulting from the process of permanently attaching chiplet **130** to the side surfaces **116** of through wafer cavity **120** while both the chiplet frontside **132** and the front surface **114** are attached temporarily to an adhesive laminate **340**. The two surfaces may be flush, such as resulting from polishing or CMP of those surfaces after removing the temporarily to an adhesive laminate **340**.

[0120] It is considered that the host wafer **110** can be vertically diced at dicing lines (shown by the

vertical bars in FIGS. 3A and 3B) along a perimeter **386** of the wafer around at least one chiplet to form a chip having the at least one chiplet and an area of the wafer surrounding the at least one chiplet.

[0121] FIGS. 3A and 3B may show how the interconnect structures **155** and **255** may sit directly on top of and touch the lateral material **360** or other dielectric material between the chiplets **130** and wafer **110**. The electrical interconnects electrically connect the passive devices of the wafer and the transistors of the chiplets. The interconnect between the chiplet and wafer may mistune the transistor, such as by introducing reactive mismatches in stability, AC signal, DC signal, impedance, frequency and/or other electronic signal matching with the input and/or output to the chiplet or a transistor on the chiplet. To avoid this mistuning, networks on the chiplet produced a robust characteristic. A benefit of any one or more of the interconnection tuning circuits **140** is that it reduces this electrical mistuning of passive devices and the transistors caused by the electrical interconnects that may impact circuit performance and reduces the uncertainty of the electrical interconnects to improve the circuit performance.

[0122] Also, chiplets **130** are preferably pre-tested to verify their functionality and/or performance. As a result, the yield of the final device **300** or diced devices is much improved over integration of component chips, in which the functionality of the component chips is not verified until after integration.

[0123] Advantageously, by allowing different electrical component chiplets **130** to be manufactured separately from each other and from wafer **110**, electronic components of those in chiplets **130** and wafer **110** can be tested separately before assembling them. In case one of the components of a certain electrical component chiplet **130** or of wafer **110** has poor fabrication yields, it is possible to replace the component to improve the fabrication yield of the assembled electrical component to produce a completed product device **300** or die thereof, having the chiplets **130** together in the cavities of the wafer **110**. For example, If an electrical component of a certain chiplet type of the chiplets **130** has poor fabrication yield, it can be rejected and replaced with a different chiplet to improve the fabrication yield of that electrical component chiplet type without spending time and money to remake the entire assembly of wafer **110**, to produce a completed product device **300** or die thereof, having the chiplets **130** and wafer **110**.

[0124] Further, because embodiments allow fabricating different electrical component ones of chiplets **130** separately from each other and from wafer **110**, all of the component types of chiplets **130** and wafer **110** do not need to be exposed to steps in the fabrication of all the different electrical component ones of chiplets **130** that could potentially damage other ones of chiplets **130** or damage wafer **110**.

[0125] Thus, embodiments can reduce manufacturing costs by using small component chips in chiplets **130** having specific features and made of exotic expensive materials, in combination with integrated circuits of other chiplets **130** and/or wafer **110** having more common features and made of cheaper common materials.

[0126] According to embodiments, chiplets **130** can include a GaN, InP or GaAs electrical component and can be fabricated on a substrate such as Si, SiGe, InP, GaAs, Alumina, or diamond. In some cases, electrical components or integrated circuits of host wafer **110** can comprise metal routing and passive components fabricated at the wafer scale. In some cases, interconnection **155** can be made using conductors made out of thin films, thick, plated interconnects, multi-layers, etc. The interconnections can for example be made using the back-end steps of a manufacturing process.

[0127] FIG. 3B is a schematic cross-sectional view of a device **301** having heterogeneous integration of a radio frequencies (RF) transistor chiplets **130** onto a wafer **310**, where the chiplets have interconnection tuning circuits **140**. Device **301** may include the one or more of the devices of any or all of FIGS. 1A-2F. As compared to wafer **110**, wafer **310** does not have cavities **120**, but instead has the chiplets mounted on a top surface of wafer **310**.

[0128] In this embodiment, the chiplets **130** are mounted on or bonded to the top surface **114** of the wafer **310** instead of being in cavities **120**. In this case, there may be no cavities for the chiplets. The interconnects **155** and **255** between the chiplets and wafer are on or above side surface **136** instead of on or above lateral material **360**. However, the interconnects may have the same mistuning noted. Here, the concepts above also apply regarding existence of interconnects **155** and **255** between the chiplets and wafer, mistuning of those interconnects, the various interconnect tuning circuits **140** for those interconnects, and the issues and advantages provided by the interconnect tuning circuits, such as noted herein.

Closing Comments

[0129] Throughout this description, the embodiments and examples shown should be considered as exemplars, rather than limitations on the apparatus and procedures disclosed or claimed. Although many of the examples presented herein involve specific combinations of method acts or system elements, it should be understood that those acts and those elements may be combined in other ways to accomplish the same objectives. With regard to flowcharts, additional and fewer steps may be taken, and the steps as shown may be combined or further refined to achieve the methods described herein. Acts, elements and features discussed only in connection with one embodiment are not intended to be excluded from a similar role in other embodiments.

[0130] As used herein, “plurality” means two or more. As used herein, a “set” of items may include one or more of such items. As used herein, whether in the written description or the claims, the terms “comprising”, “including”, “carrying”, “having”, “containing”, “involving”, and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of”, respectively, are closed or semi-closed transitional phrases with respect to claims. Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. As used herein, “and/or” means that the listed items are alternatives, but the alternatives also include any combination of the listed items.

Claims

1. An electronic assembly comprising: a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet; the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and electrical interconnects between the at least one chiplet and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein: the stabilization network includes: a first resistor between a first input of a first transistor device and a biasing node, a second resistor between a second input of a second transistor device and the biasing node, wherein the node is to a first AC ground and DC voltage that provides a first DC bias to the first and second inputs.
2. The electronic assembly of claim 1, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.
3. The electronic assembly of claim 1, wherein the interconnects electrically connect the passive devices and the transistors, wherein the at least one chiplet has passive tuning circuits to reduce electrical mistuning of the transistors caused by the electrical interconnects that degrades transistor performance and increase uncertainty of the circuit operation, and wherein each chiplet is within a

through wafer cavity of the wafer.

4. The electronic assembly of claim 1, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is 180 degrees out of phase with the first signal phase.

5. An electronic assembly comprising: a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet; the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and electrical interconnects between the chiplets and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein: the unilateralization or gain boosting network includes: at least one capacitor between a first input of the first transistor and a second output of the second transistor.

6. The electronic assembly of claim 5, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.

7. The electronic assembly of claim 5, wherein the interconnects electrically connect the passive devices and the transistors, and wherein the passive tuning circuits reduce electrical mistuning of the transistors caused by the electrical interconnects that degrades transistor performance and increase uncertainty of the circuit operation.

8. The electronic assembly of claim 5, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is 180 degrees out of phase with the first signal phase.

9. The electronic assembly of claim 5, wherein each chiplet is within a through wafer cavity of the wafer.

10. An electronic assembly comprising: a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet; the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and electrical interconnects between the chiplets and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein: the power delivery network includes: at least one capacitor between a gate of the first transistor and a first AC ground and DC voltage that provides a first DC bias to the gate of the first transistor.

11. The electronic assembly of claim 10, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.

12. The electronic assembly of claim 10, wherein the interconnects electrically connect the passive devices and the transistors, and wherein the passive tuning circuits reduce electrical mistuning of the transistors caused by the electrical interconnects that degrades transistor performance and increase uncertainty of the circuit operation.

13. The electronic assembly of claim 10, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a

differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is **180** degrees out of phase with the first signal phase.

14. The electronic assembly of claim 10, wherein each chiplet is within a through wafer cavity of the wafer.

15. An electronic assembly for heterogeneous integration of radio frequencies (RF) transistor chiplets having interconnections to or between tuning circuits, the assembly comprising: a wafer having a first circuit including passive devices for the purpose of one of tuning or matching networks for at least one chiplet; the at least one chiplet having a second circuit including at least one RF transistor or RF switch device and at least one of a stabilization network, a gain boosting network, a power delivery network, or a low-noise network for the at least one chiplet; and electrical interconnects between the chiplets and the wafer, wherein the electrical interconnects electrically connect the first circuit to the second circuit, wherein: the low-noise network includes at least one inductor between a source of the first transistor and a source of the second transistor.

16. The electronic assembly of claim 15, wherein the RF transistor device operates at and is part of one of a single-ended amplifier, differential amplifier, balanced amplifier, doherty amplifier, connected through parallel devices with distinct gate or base connections for frequency multiplication, a ring of devices for frequency mixing, or multiple devices to form an RF switch.

17. The electronic assembly of claim 15, wherein the interconnects electrically connect the passive devices and the transistors, and wherein the passive tuning circuits reduce electrical mistuning of the transistors caused by the electrical interconnects that degrades transistor performance and increase uncertainty of the circuit operation.

18. The electronic assembly of claim 15, wherein each transistor device has a gate, a source and a drain; wherein each transistor device has an input at the gate and an output at the drain; wherein a first transistor device of each chiplet is a first RF amplifier and has a first signal phase output of a differential amplifier, wherein a second transistor device of each chiplet is a second RF amplifier and has a second signal phase output of the differential amplifier, and wherein the second signal phase is 180 degrees out of phase with the first signal phase.

19. The electronic assembly of claim 15, wherein the wafer includes at least one of resistors, capacitors, inductors, through substrate vias, dielectric layers, metal layers; and wherein the wafer includes at least one layer of silicon (Si), silicon dioxide (SiO₂), silicon carbide (SiC), quartz, or Gallium Arsenide; and wherein each chiplet includes at least two RF transistors and the interconnects from the transistors to contact pads on a frontside surface of the chiplet; and wherein each chiplet is a high-end pre-fabricated transistor chiplet.

20. The electronic assembly of claim 15, wherein the wafer has front and back surfaces, through wafer cavities in the wafer formed through a plurality of areas of the top surface of the wafer, the through wafer cavities having side surfaces of the wafer; wherein the at least one chiplet is a plurality of chiplets each having a backside and a frontside disposed in the through wafer cavities, each of the chiplets having the second circuit; further comprising a lateral material between side surfaces of the chiplets and the cavity side surfaces of the wafer, wherein the lateral material bonds the side surfaces of the chiplets to the cavity side surfaces of the wafer; and wherein the electrical interconnects are over the lateral material and between the chiplets and wafer.
